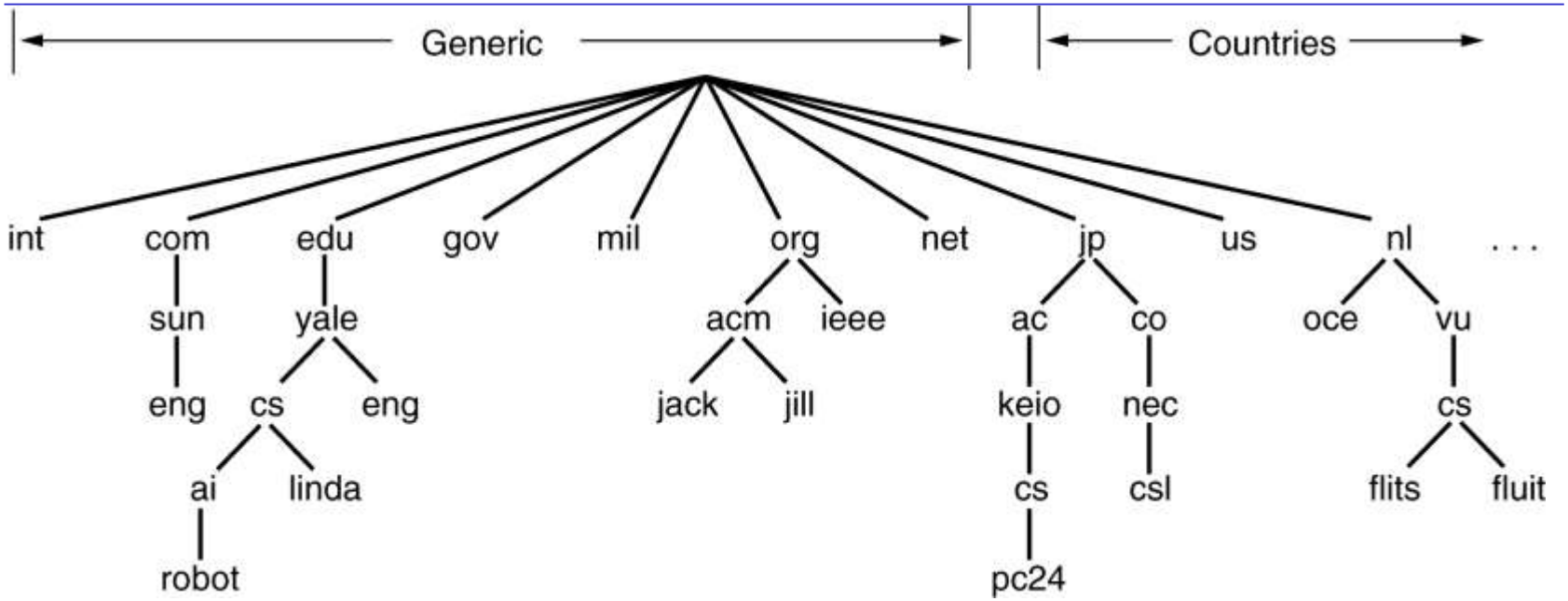
Chapter 7

The Application Layer
Domain Name System

Why?

- Computers are identified by a numerical IP address
  - and provided services by a port number
- Humans like readable, meaningful text, easy to remember
  - www.cs.ru.nl, jan@science.ru.nl, etc.
- Need for conversion between names and addresses
- This core functionality is provided by an application layer protocol: DNS

Domain based



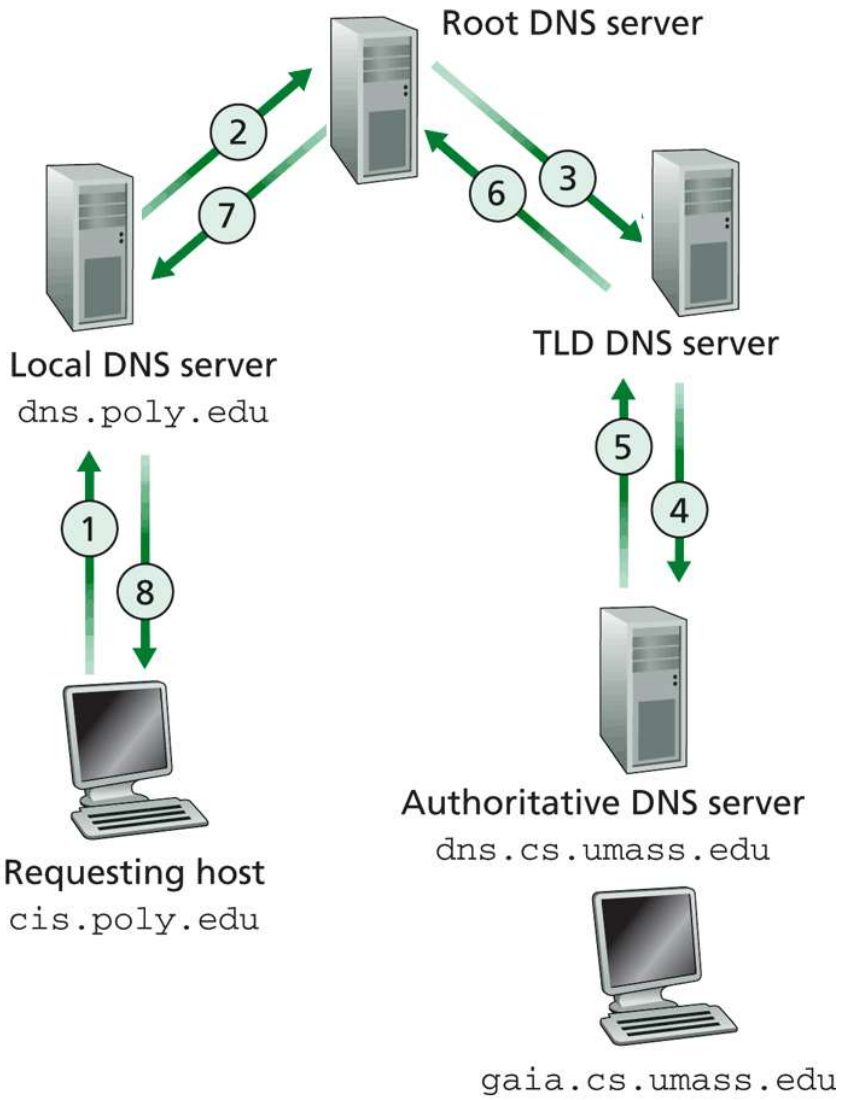
The Internet is divided into several top-level **domains**, generic (com, edu, etc.) and countries (us, nl, etc.)
Each domain is named by the path upward from it to the unnamed root. Domain names are case insensitive, each component can be up to 63 characters and the total length may not exceed 255 characters.

Distributed hierarchical database

Data for Name-IP pairs are distributed over a hierarchical organized system of servers:

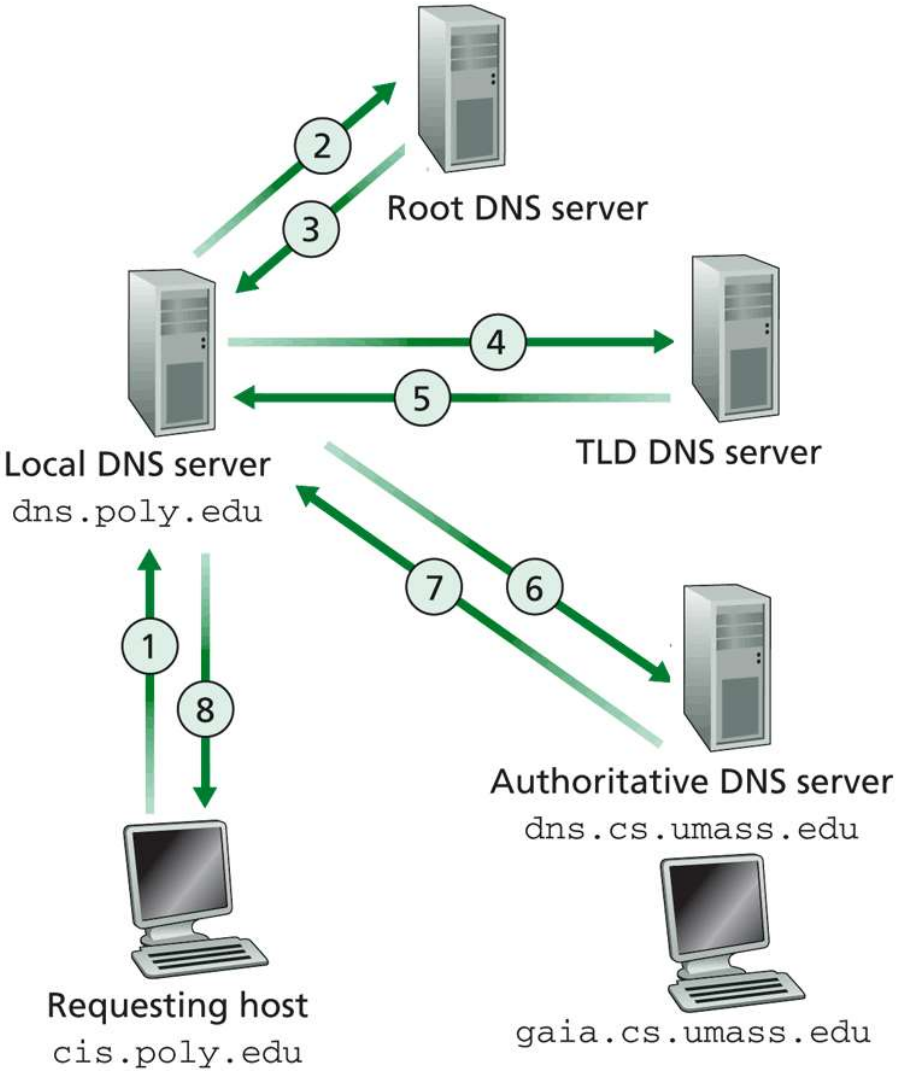
- Root DNS servers (13 now)
  - gives the IP address of the servers for the next level
  - actually each one is a cluster of servers for security and reliability reasons
- Top level domain (TLD) DNS servers
  - for each of the top level domains
  - give the IP addresses of the servers for the level below
- Authoritative DNS servers
  - one for every organization with publicly available hosts
  - contain the name-IP pairs (and more information)

Recursive lookup



- Requests go via the local DNS server first to a root DNS server, which forwards the request to the relevant TLD DNS server, which forwards it to the relevant authoritative DNS server.
- The answer goes back the same route.
- Name-IP pairs may be cached
- In case TLD only knows dns.umass.edu two more DNS messages are needed.

Iterative lookup



Requests and answers go now directly from the local DNS server into the hierarchy.
This is the most often used method.

Each DNS server caches received (name-IP) pairs for a certain amount of time.

DNS services

- host name – IP conversion
- domain name (cucg.gh) – DNS server (dns.cucg.gh) names
- host aliasing:
  - simple names for a long canonical name
  - relay1.west-coast.enterprise.com – www.enterprise.com
- mail server aliasing:
  - e.g. to use theo@enterprise.com
- load distribution over replicated servers of e.g. cnn.com
  - the DNS contains a list of IP numbers
  - the total list is returned but the order is rotated each time
  - the receiver usually takes the top of the list
  - used for web and email servers
  - recently also more complicated use for replicated servers, taking geographic distribution into account

Resource DNS records

- These are 4-tuples:
  - (Name, Value, Type, TTL)
- TTL (time to live) indicates how long it may be cached
- Types:
  - A: Name is hostname, Value the IP address
  - NS: Name is a domain, Value the name of an authoritative DNS server for it
  - CNAME: Name is an alias hostname, Value its canonical name
  - MX: Name is an alias name for a mail server, Value is its canonical name
  - there are more, like PTR (IP->hostname) and INFO (information over the host, like type and OS)

DNS messages

Identification	Flags	12 bytes
Number of questions	Number of answer RRs	
Number of authority RRs	Number of additional RRs	
Questions (variable number of questions)		Name, type fields for a query
Answers (variable number of resource records)		RRs in response to query
Authority (variable number of resource records)		Records for authoritative servers
Additional information (variable number of resource records)		Additional "helpful" info

Questions are kept when answers are send

The 16 bits identification allows to separate different requests

The flags indicate e.g. request or reply message, recursion or not, etc.

Further DNS developments

- reverse lookup via special domains, eg
  100.10.30.194.in-addr.arpa
- wildcard (*) in domain names
- ExtendedDNS : removes the length limitations
- IDNA: allowing non-ASCII characters in domain names
- DNSSEC: adding security to DNS
- new top level domains, like museum, travel, etc.
- more complicated use for replicated servers, taking geographic distribution and loads on the servers into account